

Software Test Plan (STP) Deliverables - Ritu

STP Strategy Sections

Section 4: Test Strategy & Execution Approach: Defines unit, integration, functional, usability, performance, and security testing phases across local, staging, and production environments.

Section 5: Environmental Requirements: Staging server configured with Spring Boot backend, MySQL 8.0 DB, and mock notification services.

Section 6: Entry & Exit Criteria:

Entry: 100% compilation, design sign-off, unit tests passing.

Exit: 0 Critical/High open defects, 100% execution of planned test cases.

Section 7: Defect Management: Defect tracking via Jira using standard severities (Blocker, Critical, Major, Minor).

Section 8: Test Deliverables: Test Cases, Execution Logs, Bug Reports, Traceability Matrix, Test Summary Report.

Section 12: Responsibilities: P3 owns test case creation and execution for processing, non-functional performance, usability, and reliability tests.

Section 14: Risks & Assumptions: Staging environment mirrors production capacity; API rate limits are configured for test runs.

STP §13 Traceability Matrix:

Requirement ID	Component	Test Case IDs
FR-012	Processing/Status Module	TC-PRC-01, TC-PRC-02

FR-013	Processing/Status Module	TC-PRC-03, TC-PRC-04
FR-014	Processing/Status Module	TC-PRC-05, TC-PRC-06
FR-020	Processing/Status Module	TC-PRC-07, TC-PRC-08
NFR-001	Performance	TC-PERF-01, TC-PERF-02
NFR-002	Operational / Reliability	TC-NFR-01
NFR-003	Usability	TC-NFR-02
NFR-004	Reliability	TC-NFR-03
NFR-005	Maintainability	TC-NFR-04
NFR-006	Testability	TC-NFR-05

STP Test Cases (At Least 1 Positive & 1 Negative per Requirement)

Test Case ID	Requirement ID	Description	Preconditions	Steps	Expected Result	Type

TC-PR C-01	FR-012	View complaints assigned to staff	Maintenance staff logged in; complaints assigned	1. Send GET /api/v1/staff/complaints/assigned	List of assigned complaints is displayed correctly.	Positive
TC-PR C-02	FR-012	View assigned complaints with unauthorized user	Resident user logged in	1. Send GET /api/v1/staff/complaints/assigned	HTTP 403 Forbidden returned; access denied.	Negative
TC-PR C-03	FR-013	Update complaint status using permitted state transition	Complaint in Assigned status	1. Send PATCH /status with status In Progress	Status updated to In Progress.	Positive
TC-PR C-04	FR-013	Update complaint status using illegal state transition	Complaint in Submitted status	1. Send PATCH /status with status Resolved	HTTP 400 Bad Request; state transition rejected.	Negative

TC-PR C-05	FR-014	Add remarks and resolution details upon completion	Complaint in In Progress status	1. Send PATCH /status with status Resolved and resolution remarks	Complaint state set to Resolved with saved remarks.	Positive
TC-PR C-06	FR-014	Resolve complaint without providing resolution details	Complaint in In Progress status	1. Send PATCH /status with status Resolved and empty remarks	HTTP 400 Bad Request; validation failure for missing details.	Negative
TC-PR C-07	FR-020	Close a resolved complaint via proper workflow	Complaint in Resolved status	1. Send PATCH /close request	Complaint state updated to Closed.	Positive
TC-PR C-08	FR-020	Attempt to close an unresolved complaint	Complaint in In Progress status	1. Send PATCH /close request	HTTP 400 Bad Request; complaint cannot be closed.	Negative

TC-PE RF-01	NFR-001	Verify response time under normal test load	Benchmark load running on staging	1. Send 100 concurrent complaint status requests	95% of requests complete within 10 seconds.	Positive
TC-PE RF-02	NFR-001	Verify system performance under extreme overload	Load 3x higher than peak load limit	1. Execute stress test suite	System returns 429 Rate Limit/degrades gracefully without crashing.	Negative
TC-NF R-01	NFR-002	Prevent duplicate complaints on retry	Network timeout during submission	1. Resident retries complaint submission	Duplicate record is prevented via idempotency key.	Positive

TC-NFR-02	NFR-003	Usability walkthrough for complaint submission	Resident logged in	1. Complete complaint workflow	Submission completed within ≤ 5 primary screens.	Positive
TC-NFR-03	NFR-004	Creation retry after temporary failure	Simulated server disconnection	1. Submit complaint, drop network, click retry	Single complaint record created upon recovery.	Positive
TC-NFR-04	NFR-005	Core business logic unit test coverage	Development build environment	1. Run unit test suite for core services	Unit tests pass independently with module isolation.	Positive
TC-NFR-05	NFR-006	Test core components without full system	Standalone component setup	1. Execute isolated service tests using test mocks	Service logic functions independently of full deployment.	Positive

